

HEAVY EQUIPMENT TECHNICIAN

ENGINE FUNDAMENTALS

ILM 190201-A | Second Year AIT Exam

Write-to-Learn Edition — Units A through F

Don't read it passively. Cover the right column and write the answer. If you get it wrong, write the correct answer 3 times. Do a unit, walk away, come back next day and test yourself. Repeat until you don't miss.

UNIT A — ENGINE OPERATING PRINCIPLES

The 4-Stroke Cycle

Mnemonic: ICPE — "I Can't Play Enough" — write it 3 times before moving on.

Stroke	Piston	Valves	What Happens
Intake	Down	Intake OPEN / Exhaust CLOSED	Air (or air-fuel) fills cylinder
Compression	Up	Both CLOSED	Mixture compressed — diesel 14:1–25:1
Power	Down	Both CLOSED	Fuel ignites, gases push piston
Exhaust	Up	Intake CLOSED / Exhaust OPEN	Burned gases expelled

Engine Classification — Write These

By cylinder arrangement: _____ (one row) | _____ (two rows at an angle) | _____ (flat/opposing)

Answers: Inline | V-type | Opposed/Boxer

Compression Ratio

Formula: CR = (Swept Volume + Clearance Volume) ÷ Clearance Volume

Engine Type	Typical CR
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Gasoline (SI)	8:1 – 12:1
Diesel (CI)	14:1 – 25:1
High-perf diesel	Up to 23:1

Why diesel compresses more: No spark plug — compression alone raises air temp above diesel's ignition point (~315°C). Guaranteed exam question.

Engine Displacement

Formula: $D = 0.7854 \times \text{Bore}^2 \times \text{Stroke} \times \# \text{ Cylinders}$

If bore/stroke in mm → divide by 1,000,000 for litres. If in inches → result is cubic inches.

Firing Order

Configuration	Common Firing Order
Inline 4	1-3-4-2
Inline 6 (CAT/Cummins)	1-5-3-6-2-4
V8 (generic)	1-8-4-3-6-5-7-2

Exam will ask **WHY** firing order matters: avoids consecutive cylinders, balances crank loads, minimises vibration.

UNIT B — CYLINDER BLOCK & BLOCK COMPONENTS

Block Construction

Material	Notes
Cast iron	Heavy, strong, good heat retention. Most heavy diesel.
Aluminum	Lighter, better heat dissipation. Requires steel sleeves. Softer — torque specs critical.

Cylinder Liners/Sleeves

Type	Description	Seal	Used In
Dry	Pressed into block, NO coolant contact	Press fit	Light-medium duty
Wet	Surrounded by coolant	O-rings + top flange	Heavy diesel

Cavitation erosion: Coolant pressure fluctuations create vapor bubbles that implode against liner OD → pits the outside. Prevention: correct coolant SCA concentration.

Measuring the Bore — EXAM FAVOURITE

Tool: Bore gauge (telescoping gauge + mic, OR dial bore gauge)

Take measurements at 6 positions: Top/Mid/Low × A-axis (parallel to crank) and B-axis (perpendicular).

• **Out-of-round** = A minus B at same height • **Taper** = Top minus Bottom at same axis

Deck Height / Piston Protrusion

Deck height = distance from crank centerline to block deck surface. Piston protrusion = how far crown sits above/below deck at TDC. Most diesel: piston slightly BELOW or flush. Measure with dial indicator at TDC. Determines CR and head gasket selection.

UNIT C — CRANKSHAFT, BEARINGS & RELATED COMPONENTS

Crankshaft Anatomy — Know Every Term

Term	Definition
Main Journal	Rides in main bearings — crankshaft centerline
Rod Journal (Crankpin)	Offset journal — connecting rod big end attaches here
Web	Flat plate connecting main to rod journal
Counterweight	Mass opposite rod journal — balances rotating assembly
Thrust Surface/Flange	Controls crankshaft end play (axial movement)
Fillet	Radius at journal edges — stress point, NEVER nick or grind square
Oil Hole	Drilled passage from main journal to rod journal

Write 3x: Main journal = crankshaft centerline = rides in BLOCK | Rod journal = offset = rides in CONNECTING ROD | Fillet = NEVER grind square

Crankshaft Materials

Forged steel: Strongest — heavy diesel. Narrow, sharp parting line (ID trick on exam).

Cast/nodular iron: Lighter duty. Wide, rough parting line.

Billet: Machined from solid — extreme/racing only.

Bearing Construction — THE MOST IMPORTANT PAGE

Type	Layers	Where Used
Trimetal	Steel back → copper-lead → babbitt overlay	High-load: main & rod bearings in heavy diesel
Bimetal	Steel back → aluminum alloy	Medium duty
Monoletal	Single layer (brass/bronze)	Cam bearings, bushings

Why soft babbitt overlay? Embeds contamination instead of scoring the journal. Sacrificial layer — bearing dies so the crank lives. This is by design.

Bearing Spread and Crush

Spread: Bearing half is slightly wider than bore at parting line → snaps in, stays positioned during assembly.

Crush: When halves are torqued, bearing is compressed slightly into bore → ensures full contact (heat transfer) + prevents rotation.

EXAM Q: "Why are bearing halves slightly larger than the bore?" → Spread and crush — full bore contact and prevents rotation.

Oil Clearance

Component	Typical Oil Clearance
Main bearings	0.025–0.076 mm (0.001–0.003")
Rod bearings	0.025–0.076 mm (0.001–0.003")
Cam bearings	0.025–0.076 mm (0.001–0.003")

Too little → oil can't flow, seizing. Too much → low pressure, knocking, failure.

Measuring Oil Clearance — PLASTIGAGE

1. Clean journal and bearing — NO OIL
2. Lay Plastigage strip lengthwise on journal (not across oil holes)
3. Install cap, torque to spec — DO NOT ROTATE
4. Remove cap — strip is squished
5. Measure width against scale on package
6. Wider strip = MORE clearance | Narrower = LESS clearance
7. Out of range → replace bearing or regrind journal

Common mistake: Rotating the crank with Plastigage installed — shreds the strip. Don't.

Crankshaft End Play

Measured with: Dial indicator on crank flange — pry forward, zero gauge, pry back, read movement.

Typical spec: 0.10–0.35 mm (0.004–0.014")

Thrust bearing controls end play — ONE main bearing has flanges. Usually center main.

Symptoms of excess end play: Vibration/clunk during clutch engagement, timing gear issues, front seal oil leak.

Crankshaft Inspection

Check	Tool	Reject If
Journal diameter	Outside micrometer	Below service limit
Out-of-round	Micrometer (multiple positions, same journal)	Exceeds 0.013 mm (0.0005") typically
Taper	Micrometer (front vs rear, same journal)	Exceeds 0.013 mm typically
Straightness	V-blocks + dial indicator at center main	Exceeds manufacturer limit
Surface finish	Visual + Profilometer	Scoring, cracks, rough finish

Harmonic Balancer / Vibration Damper

Dampens torsional vibration — every power stroke twists the crank. Without it, harmonics build and crack the crank.

Construction: Hub + rubber isolator + outer inertia ring. Rubber is the sacrificial element.

Inspect: Rubber deterioration; outer ring spinning relative to hub (alignment marks misaligned). Failed damper = potential crank failure + timing issues if crank sensor reads off the ring.

UNIT D — PISTONS, RINGS & CONNECTING RODS

Piston Design & Materials

Material	Notes
Cast aluminum	Most common — light, good heat transfer, expands with heat
Forged aluminum	Stronger — high-output engines
Steel crown / aluminum skirt (articulated)	Large slow-speed diesels — steel handles crown heat, aluminum skirt handles side loads

Piston-to-Wall Clearance

Measured at skirt, perpendicular to wrist pin. NOT at crown (crown runs hotter → smaller cold by design).

Typical: 0.038–0.065 mm (0.0015–0.0025") — confirm with spec sheet.

Too tight → seizes when hot. Too loose → piston slap, blow-by, oil consumption.

Piston Rings — Know All Three

Ring	Function
1 — Top compression	Seals combustion pressure. Takes most heat. Chrome-faced or nitrided.
2 — Wiper/2nd compression	Secondary seal + wipes oil off wall on downstroke. Often tapered or reverse-twist. Has TOP/UP mark — install it right-side up or it pumps oil into combustion chamber → blue smoke.
3 — Oil control	Three-piece (rails + expander spring) or single piece. Controls oil film on bore.

Exam tip: Second ring installed upside down → pumps oil UP → blue smoke + high oil consumption.

Ring End Gap

1. Insert ring into bore at specified depth
2. Push square with piston crown
3. Measure gap with feeler gauges
4. Too small → file ends (ring file ONLY — never a grinder)
5. Too large → replace ring (can't shrink it)

Too tight when hot → ring butts, seizes, breaks, scores bore. Too large → blow-by, oil burning.

Piston Pin (Wrist Pin) Types

Type	How Secured	ID It By
Full floating	Free in both rod and piston; snap rings hold axially	Snap ring grooves in piston boss
Semi-floating (rod)	Pressed into rod small end; floats in piston	No snap rings; heavy press in rod
Semi-floating (piston)	Pressed into piston; floats in rod	Frozen in piston boss

Full floating = snap rings. Missing snap ring → pin walks out → bore damage. Press fit assembly: heat piston in oil bath ~80–100°C to expand boss. Never drive cold.

Connecting Rod

Key measurements: Small end bore (wrist pin), big end bore (rod journal — measure with cap torqued), rod length center-to-center.

Inspection: Bend + twist on alignment fixture. Big end bore with bore gauge after torque.

Installation: Locating tang goes in notch. Caps are NOT interchangeable — mark before disassembly.

Torque-to-yield rod bolts: ONE-TIME USE. Replace every time. Check service manual.

UNIT E — CYLINDER HEADS & VALVE TRAIN

Cylinder Head Construction

Gasket Type	Notes
Multi-layer steel (MLS)	Modern — requires very flat deck surface
Composite	Older style — more forgiving surface finish
Fire ring/copper	High performance / racing

Head gaskets fail from overheating (warped head/deck) and thermal cycling. Always measure head flatness before reuse.

Head Resurfacing

Flatness check: Precision straightedge in multiple directions + feeler gauge. Max out-of-flat: 0.05–0.10 mm (0.002–0.004").

If you mill the head, watch for: increased CR (may exceed design), pushrod geometry changes, piston-valve interference on diesels with prechambers.

Valve Design — Angles

Face angle: 45° (some 30° intake for better flow)

Seat angle: 45° — or 46° for interference fit

Interference angle: Seat at 46°, valve face at 45° → 1° mismatch → line contact at outer edge → seals tight. As it wears, contact widens to full seat. This is the trick — know it.

Valve Margin

Thickness of valve head edge above the face. Too thin → heat damage, burning valve. Minimum typically 0.8 mm. If less than minimum after grinding → REPLACE the valve.

Valve Seat Width & Positioning

Seat width: Typically 1.0–2.0 mm intake, 1.5–2.5 mm exhaust.

Too wide → narrow with: 30° cutter (top) and 60°/75° cutter (bottom).

Lower contact band: More 30° cutter. **Raise contact band:** More 60°/75° cutter.

After seating: Prussian blue on valve → tap lightly → check contact pattern is full, no gaps.

Valve Springs

Inspect: free length vs spec | installed height vs spec (determines spring pressure) | out-of-square max 1.6 mm (1/16") | tension test at specified installed height.

Spring damper (inner spring or flat coil) prevents surge at high RPM. Install if equipped.

Valve Guides

Worn guide symptoms: Blue smoke (especially on decel), valve wobble, poor seating.

Measure: Small hole gauge + micrometer, or go/no-go gauge. Clearance = guide ID minus stem OD.

Repair options: Replace guide | knurl guide (temporary) | bronze liner insert.

Camshaft & Valve Train

Lobe Term	What It Means
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Base circle	Round section — valve is closed
Flank	Ramp up to nose (opening event)
Nose	Peak — maximum valve lift
Flank (closing)	Ramp down after nose

Cam at half speed: 4-stroke → cam sprocket is 2× crank sprocket. Write this.

Valve lift = Cam lobe lift × Rocker arm ratio (typically 1.5:1 to 1.7:1 in heavy diesel).

Valve lash: Always set hot OR cold as specified — KNOW WHICH for your engine.

Rocker Arms & Pushrods

Pushrod straightness: Roll on flat surface — wobble = bent. Replace, never straighten.

Valve adjustment sequence: Follow firing order or manufacturer sequence. Never set all valves on #1 TDC.

UNIT F — ENGINE LUBRICATION

Oil Pump Types

Type	Description	Common Use
Gear type	Two meshing gears — simple, reliable	Most common in heavy diesel
Rotor/Gerotor	Inner/outer rotor — quieter	Modern engines
Vane type	Rotating vanes — variable displacement possible	Some modern applications

Gear pump operation: Gears unmesh at inlet → low pressure draws oil in. Mesh at outlet → oil squeezed out under pressure. No check valves needed.

Oil Pressure Relief Valve

Limits maximum oil pressure. Opens when pressure exceeds spring tension → bypasses oil to sump. Located in or near pump.

Stuck open: Low oil pressure at all RPMs. **Stuck closed:** Excessive pressure → blown seals/gaskets.

Oil Flow Path — Write This From Memory

Sump → Pickup tube/strainer → Oil pump → Pressure relief valve → Oil filter → Main oil gallery → Main bearings → Crankshaft drillings → Rod bearings → Head and valve train

Oil Filter Types

Type	Bypass Valve	Notes
Full flow	Has internal bypass valve	All oil filters when clean; bypass opens on cold start or clogged filter
Bypass filter	Filters small portion continuously	Very fine filtration — polishes oil over time
Combination	Both circuits	Best filtration — heavy diesel standard

Why bypass valve in full-flow filter? Cold start — thick oil would starve the engine. Bypass opens so unfiltered oil flows rather than no oil at all. Change filter on schedule or bypass stays open permanently.

Oil Analysis — What It Tells You

Finding	Means
High iron (Fe)	Cylinder bore wear, main/rod bearing wear
High copper (Cu)	Thrust bearings, cam bearings, bushings
High aluminum (Al)	Piston wear
High silicon (Si)	Dirt ingestion — air filter failure or sealing issue
Water/coolant	Head gasket, cracked head or block, failed liner O-ring
Fuel dilution	Injector leak, incomplete combustion

EXAM STRATEGIES

What the AIT Exam Actually Tests

1. **Definitions** — name AND function of every component
2. **Procedures** — step-by-step sequences (Plastigage, valve lash, ring installation)
3. **Measurement** — what tool, what measurement, what does the result mean
4. **Specs** — they give you a measurement, you say in-spec or not (read the spec carefully)
5. **Troubleshooting** — symptom → cause → fix

Fast Troubleshooting Matrix

Symptom	Most Likely Cause
Blue smoke on startup, clears	Valve guide seals worn
Blue smoke all the time	Piston rings/bore worn, or seals
Blue smoke on decel only	Valve guide seals (oil pulled in on vacuum)
White smoke	Head gasket, cracked head, wet liner O-ring (coolant)
Black smoke	Over-fuelling, air restriction, injector issue
Knock (cold, goes away)	Piston slap — excessive piston-to-wall clearance
Knock (always, low RPM)	Rod bearing — excessive oil clearance
Knock (always, higher frequency)	Wrist pin — excessive clearance
Low oil pressure (hot)	Worn main/rod bearings, oil pump worn, thin oil
Low oil pressure (all temps)	Pressure relief valve stuck open, pickup tube blocked

Memory Tricks — Write These

ICPE = Intake, Compression, Power (explosion), Exhaust

"The bearing dies so the crank lives" — babbitt overlay is sacrificial

"Plastigage: wider = looser" — wider strip = more clearance

"Full float = snap rings" — snap ring grooves = full floating pin

"Top ring = hottest = hardest" — chrome or nitrided

"Cam at half speed" — 4-stroke: cam sprocket 2× crank sprocket

Self-Quiz — Cover All Answers and Write Yours

Do this daily the week before the exam. If you miss one, go back to that unit.

1. What is the difference between a main journal and a rod journal?

Answer: _____

2. Name the three layers of a trimetal bearing from outside to inside.

Answer: _____

3. What is 'bearing crush' and why is it important?

Answer: _____

4. How do you measure crankshaft end play and what is the typical spec range?

Answer: _____

5. What does Plastigage wider than spec tell you about oil clearance?

Answer: _____

6. What is the difference between a wet and dry cylinder liner?

Answer: _____

7. Why must the second piston ring be installed right-side up?

Answer: _____

8. What is 'valve seat interference angle' and what does it accomplish?

Answer: _____

9. Trace oil flow from the sump to a rod bearing.

Answer: _____

10. What does high silicon content in an oil sample indicate?

Answer: _____

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